

The Mathematics of RE: Paths B and C (Reverse Engineering)

1. Epistemological Foundations: The Poisson Paradox

The Poisson Paradox is a mathematical infrastructure designed to extract Expected Value (+EV) from the structural rigidities and cognitive collapses of institutional pricing algorithms, specifically those predicated on Dixon-Coles models. Rather than attempting to predict match outcomes or terminal goal counts, the framework exploits "Panic Pricing"—an irrational overshoot in live odds caused by stochastic shocks that institutional models fail to process linearly. The system identifies these discrepancies where the bookmaker's risk management prioritizes overround protection over stochastic accuracy.

The architecture operates via three distinct strategic trajectories:

- Path A (The Anchor):** A pre-match positioning on the Asian Under 2.0 line. This serves as a portfolio stabilizer, absorbing losses during low-intensity tactical matches and providing capital protection (via "Void" outcomes) to mitigate the variance of more aggressive live paths.
- Path B (The Alpha Tranche):** A dynamic live strategy that branches based on the presence or absence of early-match variance.
- Path C (The Rent Extractor):** A high-probability protocol targeting sharp Asian Under lines from the 70th minute onward to exploit terminal time decay and retail market inefficiencies.

2. The Structural Evolution: Branching Path B

The original Path B focused on capturing value in the Over 2.5 market when a match remained scoreless. However, the technical limitation of lacking high-frequency, tick-by-tick data necessitated the development of the **Reverse Engineering (RE)** branch. This allows the analyst to calculate the precise magnitude of bookmaker hyper-correction following an early goal without requiring real-time data streams.

Feature	Path B: Variance Null	Path B: RE (Variance Active)
Trigger	No goal within $t = 30$ minutes.	Early goal within $t \leq 25$ minutes.
Market Focus	Over 2.5 Goals.	Under 2.5 and Under 3.5 Goals.
Logic	Theta-capture strategy: Exploits time decay and the exodus of "dumb money."	Volatility-capture strategy: Exploits "Panic Pricing" and algorithmic overshoot.
Mathematical Goal	Capture inflated Over odds as time expires.	Capture irrationally high Under odds after a shock.

3. Mathematical Architecture of Reverse Engineering (RE)

The RE algorithm decomposes institutional pricing by inverting the bookmaker's internal logic across three stages:

1. **Extraction of the Root Parameter (λ):** The system derives the pre-match expected goals (λ) by inverting the Poisson distribution formula for the Under 2.5 Closing Line Value (CLV). *Example: A market price of 1.62 for Under 2.5 implies a total expectation of $\lambda = 2.30$.*
2. **Physiological Time Decay and Correction:** The base time-weighted decay for the expected goals at minute t is defined as: $\lambda_t = \lambda_{pre} \times \left(\frac{90 - t}{90} \right)$. To account for post-goal tactical momentum and defensive over-extension, the model applies the **Tactical Aggression Coefficient (TAC)**, a 1.05 multiplier, to derive the remaining expected goals (λ_{rem}): $\lambda_{rem} = \lambda_t \times 1.05$
3. **The Algorithmic Panic Multiplier (γ):** Institutional models frequently fail to modulate the **Bivariate Dependency (Tau - τ)** and **Decay (Rho - ρ)** parameters in real-time following a shock. This failure is captured by the Panic Multiplier (γ): $\gamma = \frac{k}{t}$. Where k is a proprietary rigidity constant and t is the minute of the goal. This multiplier identifies the "Overshoot" where the bookmaker's k -factor forces an irrational spike in Under odds to protect against a "goleada" scenario.

4. The Definitive RE Equation for Extreme Lines (Under 3.5)

The γ multiplier is most potent when applied to extreme lines such as Under 3.5. When an early goal occurs, soft bookmakers transition from protecting the Under to fearing a high-scoring blowout, causing the pricing to snap to irrationally high levels.

Panic Pricing Equation: $\text{Panic Odds} = \text{Fair Odds} \times \gamma$. By calculating the peak of the γ function, the system identifies the exact moment a line (e.g., Under 3.5) priced at 1.20 pre-match spikes to 1.70 or higher. This discrepancy represents the "Soft Bookmaker Bug," where the market price exceeds the mathematically justified Fair Odds derived from λ_{rem} .

5. Topographical Map of Market Vulnerability: The Four Zones

The efficacy of the RE trigger is categorized into four temporal zones based on the behavior of the γ function:

- The "Black Swan" event. As t approaches zero, the panic function approaches a vertical asymptote. The bookmaker experiences a total structural tilt, leading to maximum discrepancy between fair value and market price.
- The "Golden Zone" for Path B Alpha. The model identifies an **Anomalous Bounce**. While time decay is minimal, the rigidity of k produces the highest consistent +EV extraction as the bookmaker over-compensates for the early shock.
- The beginning of linear absorption. The impact of the panic multiplier remains significant, but it must now contend with a base λ_t already eroded by the first ten minutes of play.

- The limit where the RE "Bug" is deactivated. The denominator t is now large enough to justify the reduced probability of additional goals. The model returns to stochastic linearity; EV becomes marginal ($<5\%$) and the RE branch is suspended.

6. Path C: The Rent Extractor

Path C serves as the liquidity stabilizer, utilizing a "Rent Extraction" logic to capture value from the **Public Bias** prevalent in the final stages of a match.

Operational Protocol:

- **Timeline:** Active from the 70th minute onwards.
- **Trigger:** Exploits the influx of retail "dumb money" betting on late goals, which artificially inflates Under odds on sharp Asian lines.
- **Success Rate:** Historically validated at **91%**.
- **Objective:** Systematic extraction of profit from terminal Theta decay, acting as a counterbalance to the higher volatility of Paths A and B.

7. Empirical Validation and Case Studies

Simulations using 2025-2026 ground-truth data demonstrate the Edge extraction capabilities of the RE branch:

Match	Min (t)	Pre-Match CLV	Market Spike	Extracted Edge (EV)
Genoa vs. Lazio (29/09/25)	11'	1.62	3.50	+34.3%
Lecce vs. Cagliari (19/09/25)	8'	1.53	2.95	+30.0%
Torino vs. Cremonese (13/12/25)	26'	1.67	2.30	<5% (Normalizing)

8. Risk Management: Fractional Kelly and Constraint Flips

To mitigate the effects of **Overdispersion** and prevent "Gambler's Ruin," the framework employs a rigid hierarchy of financial safeguards:

1. **Volatility Paths (A & B):** Limited to **1/4 Kelly** (max 0.50 units in extreme live exposure) to manage high-variance outcomes associated with Panic Pricing.
2. **Stabilizer Path (C):** Scaled to **1/8 Kelly** for consistent rent extraction.
3. **The Constraint Flip:** An automatic logic gate triggered at a **17.89% Maximum Drawdown**. If reached, the system immediately ceases all Path B speculative activity and shifts all capital allocation to the high-success-rate Path C to recover the portfolio base.